

1.8 Classwork: Standard deviation on the GDC — Markscheme

Calculator allowed. Marks are not printed on the student sheet; they are recorded here. Total: 16 marks.

Throughout: “standard deviation” means the population value σ_x , never the sample value s_x . The two sit side by side on the One-Variable Statistics screen. Problem 1 gives $s_x = 2.58$ and Problem 2 gives $s_x = 1.24$; a candidate who quotes s_x loses the A1 for that standard deviation once only, and is followed through afterwards.

1. [Maximum mark: 9]

$$n = 10, \quad \sum x = 210, \quad \sum (x - \bar{x})^2 = 60.$$

GDC: enter the ten temperatures as a list, then *menu* $\rightarrow$ *Statistics* $\rightarrow$ *Stat Calculations* $\rightarrow$ *One-Variable Statistics*.

(a) $\bar{x} = 21^\circ\text{C}$ **A1**

$\sigma_x = \sqrt{6} = 2.4494\dots = 2.45^\circ\text{C}$ (3 s.f.) **A1**

Accept $\sqrt{6}$, 2.449, or the full 2.4494... Do not accept $s_x = 2.5819\dots$ (2.58), the sample standard deviation. The two marks are independent.

(b) Variance $= \sigma_x^2 = 6$ **A1**

Accept 6.00, or $2.45^2 = 6.0025 = 6.00$ (3 s.f.). FT: award the A1 for the square of the candidate's standard deviation in (a). A candidate who used s_x and gives 6.67 earns the A1 on FT.

(c) $\bar{F} = 1.8 \times 21 + 32$ **(M1)**

$= 69.8^\circ\text{F}$ **A1**

The M1 is for applying the formula to the mean. Converting all ten temperatures (64.4, 69.8, 66.2, ...) and running One-Variable Statistics again is a valid route to both marks. FT from the candidate's mean in (a).

(d) $\sigma_F = 1.8 \times \sigma_x = 1.8 \times 2.4494\dots$ **(M1)**

$= 4.4090\dots = 4.41^\circ\text{F}$ (3 s.f.) **A1**

Accept $1.8\sqrt{6}$, or $1.8 \times 2.45 = 4.41$. Recomputing from the ten converted values earns both marks. **The common error:** $1.8 \times 2.45 + 32 = 36.4$ — adding 32 to the standard deviation earns M0A0. FT from the candidate's standard deviation in (a); from s_x the FT answer is 4.65.

(e) Adding 32 moves every temperature, and so the mean, up by 32. **R1**

The distance of each temperature from the mean is unchanged, so the spread of the data, and therefore the standard deviation, is unchanged. **R1**

First R1 for the shift of every value (and so the mean). Second R1 for the reason the standard deviation is unaffected: distances from the mean, gaps between values, or the spread, stay the same. “Adding does not affect the standard deviation” restates the claim and earns R0 for the second mark. Accept a reason that names the multiplier: “only the $\times 1.8$ stretches the data, the $+32$ only slides it”.

2. [Maximum mark: 7]

$$n = 20 + k, \quad \sum fx = 46 + 2k.$$

$$(a) \frac{0 + 7 + 2k + 18 + 16 + 5}{2 + 7 + k + 6 + 4 + 1} = 2.2 \quad \text{that is} \quad \frac{46 + 2k}{20 + k} = 2.2 \quad \text{M1}$$

$$46 + 2k = 44 + 2.2k \Rightarrow 0.2k = 2 \Rightarrow k = 10 \quad \text{A1}$$

M1 for an equation setting $\sum fx \div \sum f$ equal to 2.2, with k in both the numerator and the denominator. An equation with k in the numerator only (for example $\frac{46+2k}{20} = 2.2$, giving $k = -1$) earns M0A0. Trial and improvement on the GDC that reaches $k = 10$ and checks $\bar{x} = 2.2$ earns both marks.

$$(b) \sigma_x = 1.2220 \dots = 1.22 \text{ goals (3 s.f.)} \quad \text{A1}$$

GDC: x values in one list, frequencies (with $k = 10$) in a second list entered as the **Frequency List**; the screen also shows $n = 30$ and $\bar{x} = 2.2$ as a check. Do not accept $s_x = 1.2429 \dots$ (1.24). FT from the candidate's k .

$$(c) \bar{x} \pm \sigma_x = 2.2 \pm 1.2220 \dots, \quad \text{that is 0.978 to 3.42} \quad \text{(M1)}$$

$$\text{Matches outside that interval: } x = 0, x = 4, x = 5: \quad 2 + 4 + 1 = 7 \text{ matches} \quad \text{A1}$$

M1 for at least one of $\bar{x} - \sigma_x$ and $\bar{x} + \sigma_x$, seen or implied. Counting one tail only (5 matches above, or 2 below) earns M1A0. Counting the matches inside the interval (23) earns M1A0. FT from the candidate's standard deviation in (b); s_x gives the same interval in whole numbers and the same 7.

$$(d) \text{ The mean decreases, because 2 is less than the mean of 2.2.} \quad \text{A1}$$

The standard deviation decreases, because 2 is very close to the mean (well within one standard deviation of it), so the new match adds less than the typical squared distance from the mean. **R1**

New values: $\bar{x} = \frac{68}{31} = 2.19$ and $\sigma_x = 1.20$ (3 s.f.). Values are not required, and a candidate who recomputes on the GDC and compares ($2.19 < 2.2$, $1.20 < 1.22$) has given a valid reason for both marks. A1 for the mean decreasing with its reason. R1 for the standard deviation decreasing with a reason that refers to 2 being close to the mean. "Stays the same, because 2 is a typical value" earns R0. A correct direction with no reason earns no mark for that statistic.